

Self-Hosted Trading Infrastructure on openSUSE

From autonomous crypto agents to the coming world of tokenized stocks

This is not financial advice

WHAT WE'LL COVER

Two halves of one idea



PART 1

The system today

1. The core idea: Linux as an always-on host
2. Why self-hosted beats the black box
3. Architecture: RSI, spot trades, systemd
4. Configuration by environment variable
5. Conservative vs. aggressive tuning
6. Leap, Tumbleweed & MicroOS
7. Beyond trading and the risks



PART 2

Where it goes next

1. Tokenized stocks & real-world assets
2. Why the same agent fits perfectly
3. Future use cases for programmable portfolios
4. The TradFi ↔ DeFi convergence
5. Hard problems still to solve
6. How you could start building today

THE CORE IDEA

Linux as a home for always-on autonomous services

No flashy UI.

No proprietary cloud.

No opaque black box.

No monthly fee.

Instead: a service that runs reliably 24/7, executing predefined policies in response to market signals.



Continuous by design

Runs quietly around the clock, sometimes doing nothing at all, which is the point.



Autonomous, yet accountable

Operates with minimal human intervention while keeping every decision inspectable.



Open tooling, auditable logic

Deploy and audit your own automation without surrendering control to a platform.

WHY SELF-HOST

Transparency over convenience

The trading app / SaaS model

- Closed platform you must trust blindly
- Opaque logic; you can't see why it acted
- Recurring fees for access
- No rollback, no fine-tuning, no control
- Your strategy lives on someone else's server

The self-hosted model (container or)

- ✓ Configurations are easy and human-readable
- ✓ Logs are readable via journalctl
- ✓ Services managed cleanly through systemd
- ✓ Inspect, roll back or fine-tune at will
- ✓ You never surrender control to a closed platform

ARCHITECTURE AT A GLANCE

A quiet loop: sense → decide → act



01

Check price

Poll recent candle / price data on an interval



02

Compute RSI

Derive the Relative Strength Index of recent candles



03

Apply policy

Below threshold → consider buy · above → consider sell



04

Spot trade

Execute in spot markets only - no leverage

RUNS ON



Node.js service



systemd (user unit)



journald logging

THE SIGNAL

How the decision is made: RSI

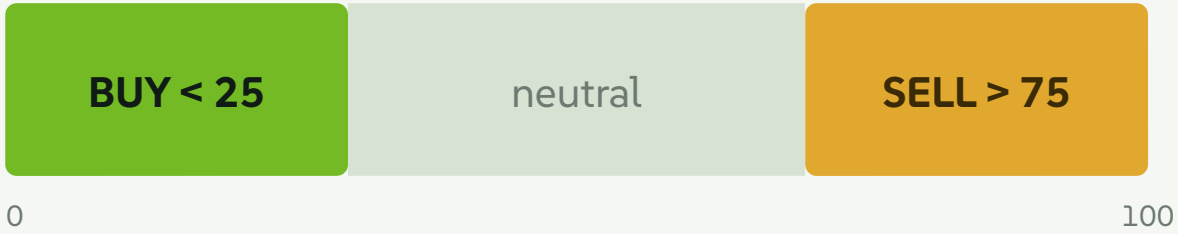
The Relative Strength Index measures the momentum of recent price moves on a 0–100 scale.

Low RSI → asset looks oversold → the agent considers buying.

High RSI →
asset looks overbought → the agent considers selling.

Spot markets only · no leverage · not high-frequency

RSI SCALE



RSI_PERIOD = 14

Fourteen candles is a common default for 2-hour charts, WHICH balances responsiveness against noise. Shorter periods react faster but fire more signals; longer periods smooth fluctuations and trade less often.

OPERATIONS

Running the agent on openSUSE

zsh — jup-bot

```
# 1 · install the runtime
$ sudo zypper in nodejs

# 2 · start manually
$ node --max-old-space-size=8192 dist/bot.js

# 3 · run as a background service (.env with chmod 600)
$ systemctl --user enable --now jup-bot.service

# 4 · follow the logs
$ journalctl --user -u jup-bot.service -f
```

User-level systemd unit

Managed like any other service as in enable, start, restart, inspect.

Same on every flavor

Works identically on Leap, Tumbleweed and MicroOS.

Safe updates on MicroOS

Transactional updates roll back automatically if one ever breaks the runtime.

CONFIGURATION

Everything is an environment variable

All trading behavior is controlled through environment variables; not hard-coded values. Changes can be tested, reverted or tuned without recompiling. Restart the service and the new policy is live.



14

RSI_PERIOD

Candles used to calculate RSI



25

RSI_BUY

Lower threshold that triggers a buy



75

RSI_SELL

Upper threshold that triggers a sell



0.40

TRADE_FRACTION

Share of balance committed per trade



0.20

MIN_SOL_RESERVE

Reserve kept back, never traded



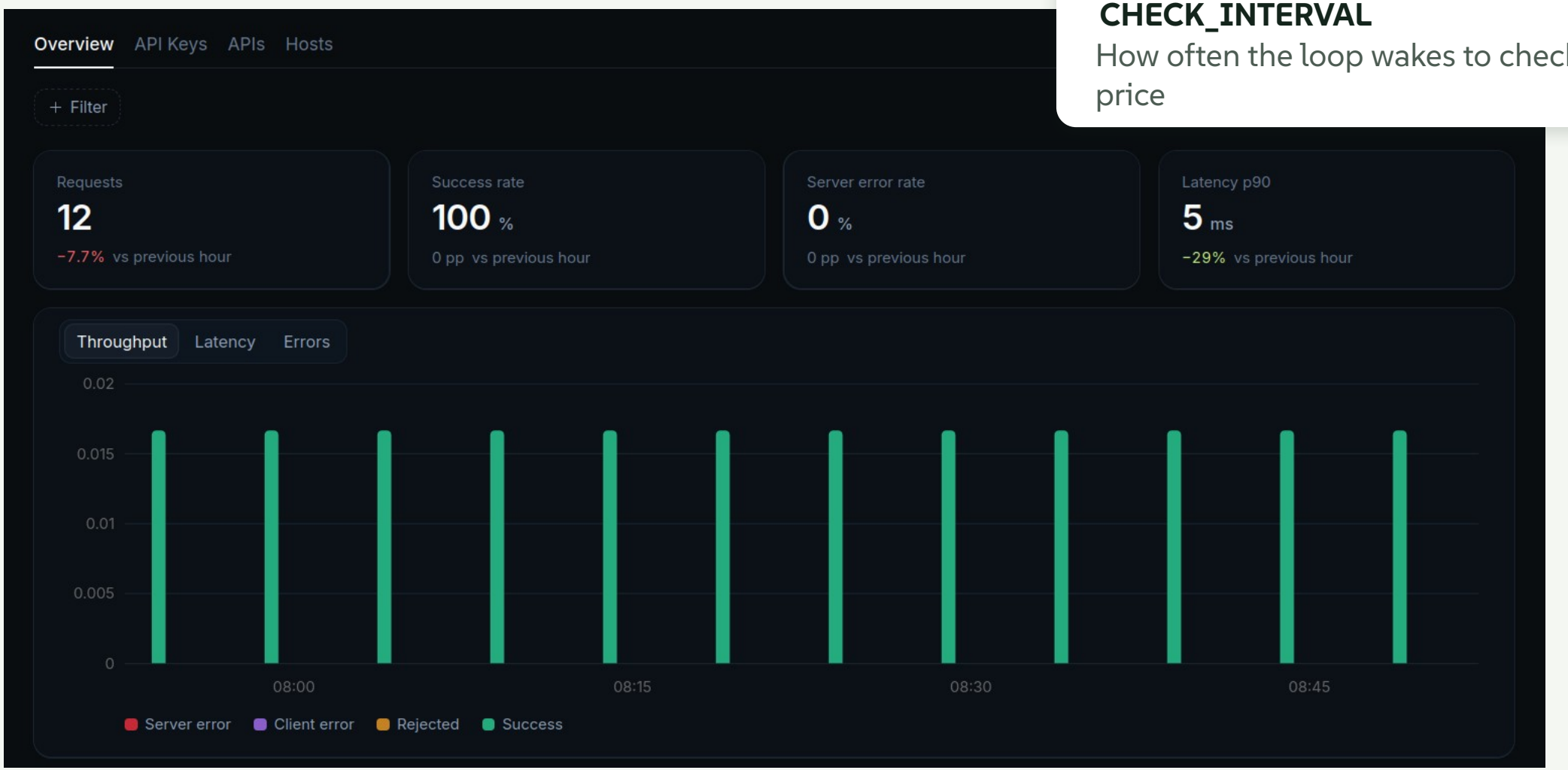
900s

CHECK_INTERVAL

How often the loop wakes to check price

TUNING

My CHECK_INTERVAL



900s

CHECK_INTERVAL

How often the loop wakes to check price

TUNING

Same code, two personalities

Conservative

Trades less · holds more back · waits for stronger signals

TRADE_FRACTION	= 0.10
MIN_SOL_RESERVE	= 0.20
RSI_BUY	= 30
RSI_SELL	= 70
CHECK_INTERVAL	= 1800
COOLDOWN	= 7200

Aggressive

Trades more · smaller reserve · reacts to gentler moves

TRADE_FRACTION	= 0.80
MIN_SOL_RESERVE	= 0.05
RSI_BUY	= 30
RSI_SELL	= 70
CHECK_INTERVAL	= 300
COOLDOWN	= 900

CHOOSE YOUR BASE

One agent, three openSUSE flavors



Leap

Enterprise-grade stability

Slower change and long support cycles with a clean upgrade path to SUSE Linux Enterprise Server when you want it.



Tumbleweed

Up-to-date & fast-moving

Current kernels and libraries and is ideal for developers working against fast-moving blockchain and finance SDKs.



MicroOS

Immutable & self-healing

Transactional updates and an immutable filesystem. It is built for unattended edge services where rollback beats manual tweaking.

GENERALIZING

The trading bot is just one shape

The architecture applies equally to any continuous, policy-driven workload. Swap the signal and the action - keep the systemd-supervised loop.



Telemetry processing

Continuously ingest and transform streams of sensor or system data.



Monitoring systems

Watch endpoints and services, react to state changes automatically.



API polling services

Wake on an interval, query external APIs, act on what changed.



Data aggregation pipelines

Collect from many sources into one consolidated, queryable store.

READ THIS TWICE

Risk & responsibility

A conservative posture reduces risk — it does not eliminate it. Users assume full responsibility for both market exposure and technical failure modes.

Market risk is real

Prices move against you. Spot-only and no leverage bound the damage, but losses are still possible.

Technical failure modes

Crashes, bad configs, API outages and bugs are yours to own, monitor and recover from.

Not financial advice

This is a technology use case, not a recommendation. Nothing here tells you what to trade.



PART 2

The Future: Tokenized Assets

What happens when the very same agent stops trading crypto and it starts trading tokenized stocks, bonds and funds.



TRADITIONAL SYSTEM IS AT RISK

Because all assets on your bank account are numbers in a data base

DEFINITIONS

What are tokenized stocks & RWAs?



Real-World Assets (RWAs)

Traditional financial assets represented as programmable tokens on a blockchain — a digital wrapper around a claim on something real.

Same claim, new rails: 24/7 settlement, fractional ownership, and programmatic access via API.



Equities

Tokenized shares in public companies



Treasuries & bonds

On-chain government and corporate debt



Funds & ETFs

Index and money-market funds as tokens



Commodities

Gold, metals and other stores of value

THE BRIDGE

Why the same agent fits perfectly



TODAY

RSI agent trades a digital asset

same loop, new asset



TOMORROW

Same agent trades tokenized AAPL, ATG...



Markets never close

Tokenized assets can trade 24/7 — exactly what an always-on agent expects.



API-native access

Programmatic order flow is built in, no brokerage GUI in the loop.



Policy unchanged

RSI, thresholds, reserves and cooldowns all still apply verbatim.

WHAT BECOMES POSSIBLE

Programmable portfolios: future use cases

Automated rebalancing

Hold target weights across tokenized equities and bonds; the agent trims and tops up as prices drift.

Dollar-cost averaging

Buy fixed fractions on a schedule, fully unattended, into index tokens or blue-chip shares.

Dividend reinvestment

On-chain dividends detected and automatically redeployed the moment they settle.

Cross-asset strategies

Rotate between tokenized stocks, treasuries and commodities on a single, unified policy.

Tax-loss harvesting

Rules that realize losses within your constraints, logged transparently for your records.

Yield laddering

Roll tokenized treasuries across maturities to keep idle cash productively deployed.

THE BIG PICTURE

TradFi and DeFi are converging

Two worlds, one API

Traditional assets gain blockchain rails; on-chain tooling gains real-world value to work with.

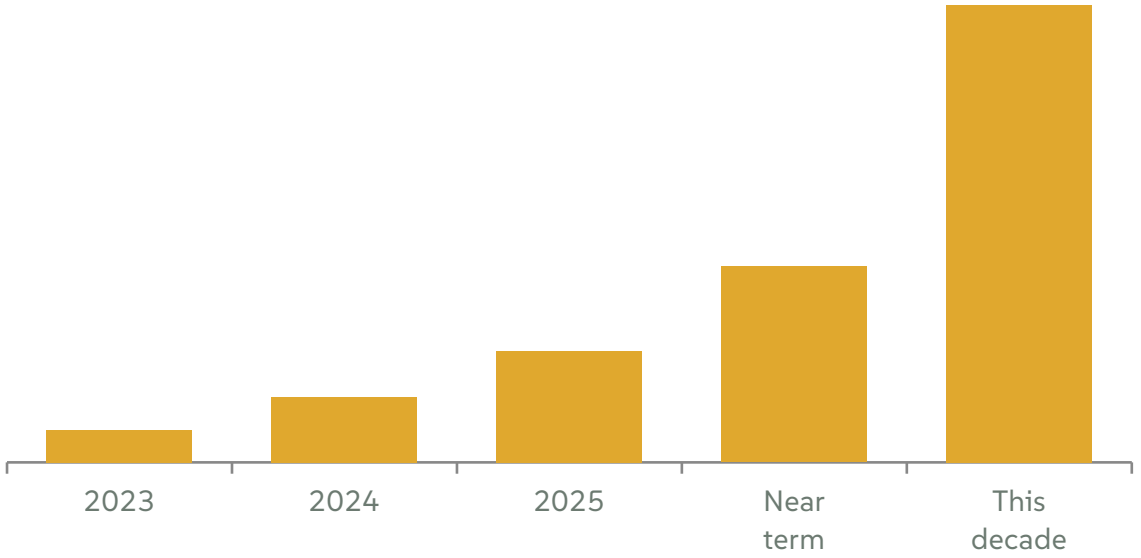
Always-on by default

The 9:30-to-4 trading day gives way to continuous, global, programmable markets.

Self-custody & control

The transparency ethos of self-hosting extends from your Linux box to your portfolio.

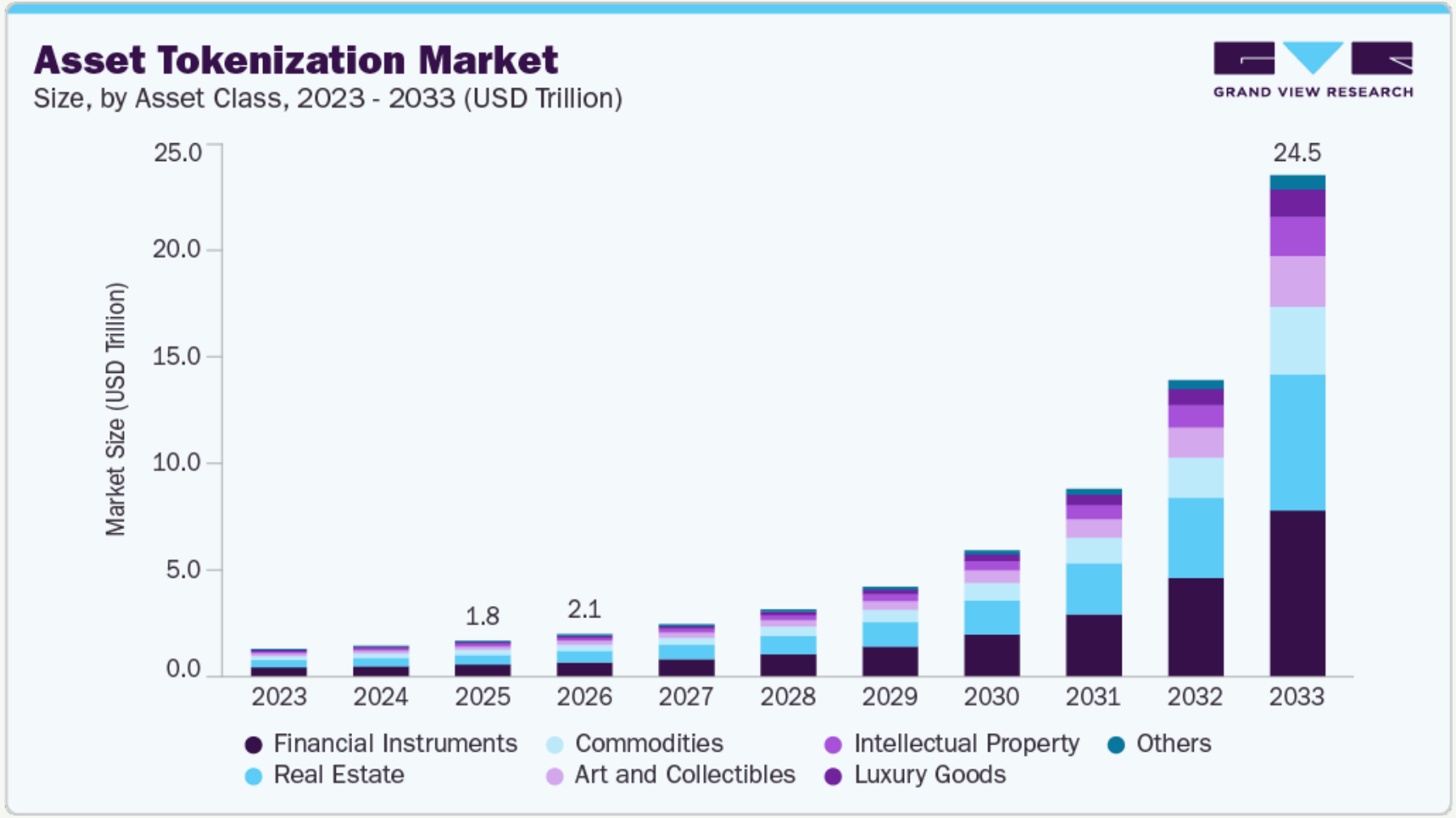
Tokenized RWAs — illustrative growth trajectory



Illustrative only; analyst estimates of the tokenized-asset market vary widely and run into the trillions. Shown to convey direction, not a forecast.

THE BIG PICTURE

TradFi and DeFi are converging



https://www.grandviewresearch.com/industry-analysis/asset-tokenization-market-report

STAYING HONEST

Hard problems still to solve

Regulation & compliance

Tokenized securities sit under real securities law. Rules differ by jurisdiction and are still evolving. **Current status is Europe is that the tokenized security is representational.**

Custody & key security

Self-custody means you own the keys, and the catastrophic risk of losing them. No support desk to call.

Oracles & price feeds

An agent is only as good as its data. Manipulated or stale price oracles lead to bad automated decisions.

Liquidity & spreads

Thin markets mean slippage. Automated orders can move price against themselves in illiquid tokens.

Smart-contract risk

Bugs in the underlying token or venue contracts can be exploited, code you don't control.

Automation blind spots

An unattended agent will faithfully execute a flawed policy at scale. Guardrails are non-negotiable.

FROM HERE

How you could start building today

1 Stand up the base

Install openSUSE — MicroOS for unattended, Tumbleweed for development — and Node.js via zypper.

2 Build the loop

Write a small sense-decide-act service. Keep every parameter in environment variables from day one.

3 Supervise with systemd

Run it as a user unit. Tail it with journalctl. Add restart policies and health checks.

4 Paper-trade first

Simulate against live prices with no real orders. Prove the policy before a single cent is at risk.

5 Tune in stages

Start conservative. Adjust thresholds and fractions one at a time, observing each change.

6 Watch the RWA space

Track tokenized-asset venues and regulation in your jurisdiction — then extend the same agent.

THE AI STORY

Do you think it will survive AI exploitation?

THE REAL STORY

The bot watches charts, but the story is control for the future